MIL-I-21203A(AS)
Amendment-2
15 November 1969
Superseding
Amendment-1
15 October 1963

MILITARY SPECIFICATION

INDICATOR, ATTITUDE, SPHERICAL, DIRECT READING,

FAST ERECTING, ELECTRIC GYRO

This Amendment forms part of Military Specification MIL-I-21203A(WEP) of 1 August 1962 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this Amendment MIL-I-21203A(WEP) is changed to MIL-I-21203A(AS) and throughout the specification Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

*Page 2, Paragraph 2.1, under Specifications Military: Add

"MIL-C-14806 Reflective Coating
MIL-G-81559 Packaging, Packing, Preservation and Marking
MIL-B-81744 Barrier Coating Solution and Solvent"

*Page 2, Paragraph 2.1, under Standards-Military: Add

"MIL-STD-1334 Barrier Coating Process for Anti-Friction Bearings"

*Page 5, Paragraph 3.4.4.3: Delete and Substitute

"3.4.4.3 Demand Erection Device - A pushbutton shall be provided on the face of the indicator to actuate a fast erection system when the instrument is operating. The push button shall be required to be held down in the fast erect position to bring the gyro to its gravity position from any random position in pitch and roll. The push button shall be marked "push for fast erection."

3.4.4.3.1 Caging - (As an alternate to paragraph 3.4.4.3, above) Means shall be incorporated to cage the gyroscope by pulling the pitch trim knob. With the knob fully extended and rotated to the extreme clockwise position, the gyroscope shall remain caged and the knob shall remain locked in the extended position. When the knob is pulled and power is applied, the power flag shall fail to appear.

FSC 6610

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Page 5, Paragraph 3.5.3, first sentence: Delete and Substitute

"The electrical connectors shall be of the type used for a hermetically sealed case and shall mate with connector types MS-3102-20-29P and MS-3102-18-1P conforming to specification MIL-C-5015."

*Page 7, after Paragraph 3.6.1: Add

"3.6.1.1 Reflective Coating - All reflecting glass surfaces of the cover glass and lighting wedge shall be covered with a reflection reduced coating that meets the requirements of MIL-C-14806."

*Page 8, Paragraph 3.7: Delete and Substitute

"3.7 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.7.1 Operation - The indicator shall provide accurate visual presentation of aircraft displacement in the roll and pitch axes which are referenced to gravity, without requiring inputs from any external source other than power supply.

3.7.1.1 Accuracy -

3.7.1.1.1 Standard Conditions - The indicator, when connected and operated at standard conditions, shall provide a pitch and roll accuracy of 0.50 degree minimum.

3.7.1.1.2 Longitudinal Acceleration - The indicator, when subjected to an 8g acceleration for 4 seconds along the roll axis, shall provide a pitch and roll accuracy of 2 degrees minimum.

3.7.1.1.3 Coordinated Turns - The indicator, when subjected to coordinated turns with rates of turn between 90 and 1500 degrees per minute and bank angles of 20 to 60 degrees, shall provide a pitch and roll accuracy of 2 degrees minimum.

3.7.1.1.4 Uncoordinated Turns - The indicator, when subjected to uncoordinated turns (skid turns) of less than 5 degrees per minute at air speeds not greater than 400 knots for a 10-minute duration, shall provide a pitch and roll accuracy of 2 degrees minimum."

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*Page 8, Paragraph 3.8.6: Delete and Substitute

"3.8.6 Lubrication - The lubricating oil used shall meet the performance requirements of specification MIL-L-6085 except that it shall conform with Navy Research Laboratory polyol-ester lubricant NRL MB-20-B.

*Page 12: Add

"3.8.12 Barrier Film Bearing Coating - All bearings lubricated in accordance with requirements of paragraph 3.8.6 shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with Specification MIL-B-81744. The bearings shall be barrier film processed either by a ball bearing manufacturer or a designated government facility."

*Page 13, Paragraph 4.4. and all Subparagraphs: Delete and Substitute

"4.4. Reliability Production Acceptance Phase - Reliability Production Acceptance Testing shall be conducted through a production burn-in. Each equipment produced but not submitted to the Qualification Test shall be operated for 150 operating hours burn-in, and the test data (operating hours and failures) shall be compared to the Accept/Reject Criteria of Test Plan V of MIL-STD-781. Delivery of equipment shall not be held up to await a decision under the test plan. In event an Accept Decision is reached, the test duration of each equipment shall be reduced to 100 operating hours. At any time that the current total of test hours and test failures plotted on Test Plan V curves show a Reject Situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a Reject Situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a rejection decision. For the Production Acceptance Phase, Test Level F shall be used, except that the temperature extremes shall be +55°C and -44°C, and vibration shall be 0.5G."

*Page 21, Paragraph 4.6.26: Delete and Substitute

"4.6.2 Vibration Failure Test - The indicator shall be tested in accordance with MIL-T-5422 using Curve III except that the frequency range shall be extended to 2000 cps and the amplitude below 350 cps shall be limited to 2 g's. The indicator shall be hard mounted to the vibration table during tests. At the conclusion of the tests the equipment shall meet the requirements of the individual tests. It shall also be demonstrated that the gimbal vibration response has not caused false brinelling of bearings as a result of this test. The requirements of paragraphs 3.8.6 and 3.8.12 shall be incorporated in the equipment for this test. After 50 hours testing, fretting corrosion shall not be present."

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Page 21: Add

"4.6.33 Impact Shock - The indicator shall be tested in accordance with the shock test of MIL-T-5422. At the end of this test, the equipment shall be subjected to and shall meet the requirements of the individual tests.

4.6.34 Shock Test- The indicator shall be tested in accordance with the shock test of MIL-T-5422, with power applied. The equipment shall then be subjected to and meet the requirements of the individual tests."

*Pages 21 and 22, Paragraphs 5.1, 5.2, 5.2.1, 5.2.2, 5.3 and 5.3.1: Delete and Substitute

"5.1 Packaging, Packing, Preservation and Marking - Shall be in accordance with MIL-G-81559."

An asterisk preceding a page number denotes a change to be made to the specification which did not appear in Amendment-1.

Project Number 6610-N047

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